Tutorial Sheet 13 Solutions

Discrete Mathematical Structures

Question 1

(a) Check if the following groups are cyclic.

A group is cyclic if there exists a generator element whose powers (under multi.) or multiples (under addition) produce the entire set.

i. $(\{1, \omega, \omega^2\}, \cdot)$
Let the generator be ω. We evaluate its powers:

$$\omega^1 = \omega$$
$$\omega^2 = \omega^2$$
$$\omega^3 = 1$$

The powers of ω generate all elements in the set. The group is **cyclic** (generators: ω, ω^2).

ii. $(\{1, -1, i, -i\}, \times)$
Let the generator be i (the imaginary unit). We evaluate its powers:

$$i^1 = i$$
$$i^2 = -1$$
$$i^3 = -i$$
$$i^4 = 1$$

The powers of i generate all elements in the set. The group is **cyclic** (generators: $i, -i$).

iii. $(\mathbb{Z}_4, +)$
For the additive group $\mathbb{Z}_4 = \{0, 1, 2, 3\}$, let the generator be 1. We evaluate its successive multiples modulo 4:

$$1 \cdot 1 \equiv 1 \pmod 4$$
$$2 \cdot 1 = 1 + 1 \equiv 2 \pmod 4$$
$$3 \cdot 1 = 1 + 1 + 1 \equiv 3 \pmod 4$$
$$4 \cdot 1 = 1 + 1 + 1 + 1 \equiv 0 \pmod 4$$

The multiples of 1 generate the entire set. The group is **cyclic** (generators: $1, 3$).

Sol. Q1 (b)

Problem Statement: Prove that the set of integers $S = \{0, 1, 2, 3, 4, 5, 6\}$ forms a Ring under addition modulo 7 ($+_7$) and multiplication modulo 7 ($\times_7$). This set is denoted as $\mathbb{Z}_7$.

1. Addition Modulo 7 ($+_7$): For $(\mathbb{Z}_7, +_7)$ to be an **Abelian Group**, it must satisfy closure, associativity, identity, inverse, and commutativity.

$+_7$	**0**	**1**	**2**	**3**	**4**	**5**	**6**
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

Table 1: Addition Table for $\mathbb{Z}_7$

- **Closure:** For any $a, b \in \mathbb{Z}_7$, the result of $a +_7 b$ is always an element of the set $\{0, 1, 2, 3, 4, 5, 6\}$ as in Table 1.
- **Identity:** $\exists \quad 0 \in \mathbb{Z}_7$ such that for every $a \in \mathbb{Z}_7$, $a +_7 0 = 0 +_7 a = a$. Thus, 0 is the additive identity.
- **Inverses:** $\forall a \in \mathbb{Z}_7, \exists -a \in \mathbb{Z}_7$ such that $a +_7 (-a) = 0$. Examples: $1 +_7 6 = 0$, $2 +_7 5 = 0$, and $3 +_7 4 = 0$.
- **Commutativity:** For any $a, b \in \mathbb{Z}_7$, $a +_7 b = b +_7 a$. As we can see the table is symmetric across the main diagonal.

2. Multiplication Modulo 7 ($\times_7$): For $(\mathbb{Z}_7, \times_7)$ to be a **Semigroup**, it must be closed and associative.

$\times_7$	**0**	**1**	**2**	**3**	**4**	**5**	**6**
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
5	0	5	3	1	6	4	2
6	0	6	5	4	3	2	1

Table 2: Multiplication Table for $\mathbb{Z}_7$

- **Closure:** $\forall a, b \in \mathbb{Z}_7, (a \times_7 b) \in \mathbb{Z}_7$.
- **Associativity:** $a \times_7 (b \times_7 c) = (a \times_7 b) \times_7 c, \forall a, b, c \in \mathbb{Z}_7$.

Since $(\mathbb{Z}_7, +_7)$ is an Abelian group, $(\mathbb{Z}_7, \times_7)$ is a semigroup, and multiplication distributes over addition, the structure $(\mathbb{Z}_7, +_7, \times_7)$ is a **Ring**.

Question 2

For a set $A = \{1, 2, 3\}$, check if the following relations are POSETS.

A relation is a Partially Ordered Set (POSET) if it is strictly Reflexive, Antisymmetric, and Transitive.

(i)$R1 = \{(1, 1), (2, 2), (3, 3)\}$

Reflexive: Yes (contains (a, a) for all $a \in A$).

Antisymmetric: Yes.

Transitive: Yes.

$R1$ **is a POSET.**

(ii)$R2 = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$

Reflexive: Yes.

Antisymmetric: No. It contains both $(1, 2)$ and $(2, 1)$, which would require $1 = 2$ for antisymmetry to hold.

$R2$ **is NOT a POSET.**

(iii)$R3 = \emptyset$

Reflexive: No. It lacks the required self-pairs $(1, 1), (2, 2), (3, 3)$.

$R3$ **is NOT a POSET.**

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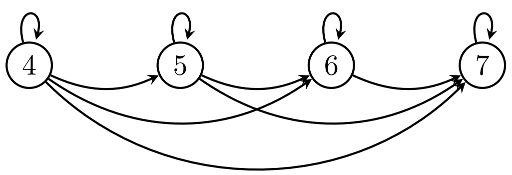
Question 3

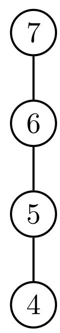
Ans.3 Consider the set $A = \{4, 5, 6, 7\}$. Let R be the relation $\leq$ on A. Draw the directed graph and the Hasse diagram of R.

The relation set is $R = \{(4, 4), (4, 5), (4, 6), (4, 7), (5, 5), (5, 6), (5, 7), (6, 6), (6, 7), (7, 7)\}$.

1. Directed Graph:
Self-loops exist on every node (reflexivity). Directional arrows point from smaller numbers to larger numbers.



2. Hasse Diagram:
Removing self-loops and redundant transitive edges results in a linear, upward-flowing hierarchy.

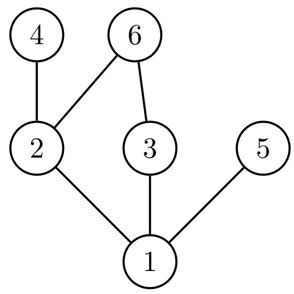


Question 4

Let $X = \{1, 2, 3, 4, 5, 6\}$. Draw Hasse diagram for $(X, |)$, where $|$ represents the 'divides' relation.

Nodes are arranged such that an upward path exists from a to b only if a divides b.

- **Level 1 (Bottom):** 1 (Divides all elements).
- **Level 2:** $2, 3, 5$ (Prime numbers, divisible only by 1). Note: 5 acts as a maximal element on its own branch.
- **Level 3 (Top):** 4 (divisible by 2) and 6 (divisible by 2 and 3).



Solution for Question 5

Definitions: Let $(P, \leq)$ be a partially ordered set and $S \subseteq P$.

Maximal Element: $m \in S$ is maximal if $\nexists x \in S$ such that $m < x$.

Greatest Element: $g \in S$ is the greatest element if $\forall x \in S, x \leq g$.

Upper Bound (UB): $u \in P$ is an upper bound of S if $\forall s \in S, s \leq u$.

Supremum (LUB): $x = \sup(S)$ if $x \in UB(S)$ and $x \leq u$ for all $u \in UB(S)$.

A poset (P, $\leq$) is a **lattice** if for all a, b$\in$P : a$\wedge$b i.e. glb (a, b) exists and a$\vee$b i.e. lub (a, b) exists

Solution:

Subset	**UB**	**LB**	**LUB**	**GLB**
$\{d, k, f\}$	$\{k, l, m\}$	NIL	k	NIL
$\{b, h, f\}$	$\{k, l, m\}$	NIL	k	NIL
$\{d\}$	$\{d, h, i, j, k, l, m\}$	$\{a, b, d\}$	d	d
$\{a, b, c\}$	$\{k, l, m\}$	NIL	k	NIL
$\{l, m\}$	NIL	$\{a, b, c, d, e, f, g, h, k\}$	NIL	k

Maximal Element={l, m}
Minimal Element={a, b, c}
Greatest Element: {}
Least Element: {}

Question 6: Lattice Verification

Mathematical Logic: A poset $(P, \leq)$ is a lattice if and only if for every pair of elements $x, y \in P$, both the *Join* ($x \vee y$, the least upper bound) and the *Meet* ($x \wedge y$, the greatest lower bound) exist and are unique.

(a) Is a Lattice.
In this poset, for any arbitrary pair $\{x, y\}$, there is a unique $z = \sup\{x, y\}$ and $w = \inf\{x, y\}$. The vertical structure ensures that paths converge to a single supremum and infimum for all pairs.

(b) Is NOT a Lattice.
The poset fails the lattice criteria because the pair $\{b, c\}$ does not have a **Least Upper Bound (LUB)**.

Proof by Contradiction:

- The set of Upper Bounds (UB) for $\{b, c\}$ is $\{d, e, f\}$.
- By definition, an element u is the LUB if $u \leq x$ for all other upper bounds x.
- However, in this diagram, $d \parallel e$ (elements d and e are incomparable). Since neither $d \leq e$ nor $e \leq d$, there is no unique *least* element in the set of upper bounds.